

Supercluster — Full Production Scroll Script

Status: Production Draft v1

Global System Constants

Particle Field

Property	Value
Total particles	800k desktop / 200k mobile
Chapter particle pool	10-18% of total field per chapter
Ambient field	Remaining particles: slow drift + filament formation throughout every chapter
Base particle color	Deep cobalt (● #0A1628) with gold trace filaments (● #C9A84C)
Star points	~0.1% of field — fixed bright white, never recruited for artifacts

Particle Behavior Definitions

- **Condense** — particles drift toward target positions; local density increases
- **Route** — particles form directional filaments with visible flow direction
- **Carve** — particles concentrate at silhouette boundary; interior remains void
- **Pulse** — brightness oscillates at beat frequency (1-3 Hz)
- **Shear** — particles accelerate outward from artifact center on chapter end

- **Rejoin** — sheared particles decelerate and merge back into ambient field drift

Scroll Mechanics

Property	Value
Each dot	One scroll snap unit
Particle morph duration	800–1200ms
Camera leads morph by	200ms
Proof card fade in	400ms after morph begins
Proof card fade out	200ms before next beat begins
Chapter transition	Camera Shear → Rejoin → travel → arrive at next node

Camera Constants

- **Resting:** oblique top-down, ~55° elevation, slow ambient drift
- **Chapter entry:** drifts toward density node over 2–3s as first particles begin to Condense
- **Beat transitions:** micro-adjustment $\leq 15^\circ$ rotation over 1s, holds during proof card read
- **Chapter exit:** pulls back ~0.5 FOV, then reorients toward next node while Rejoin plays

Chapter Color Identities

Chapter	Primary	Secondary	Emotion
MIRA	Gold  #C9A84C	Warm amber  #FF9500	Signal, warmth, voice
AIDEN	Cyan  #00D4FF	Ice blue  #A8D8FF	Data, conversation, clarity
VANGUARD	Red-violet  #8B2FC9	Pass-green  #00FF7F	Test, fail, reroute
INSPECTION	Scan blue  #A8D8FF	Defect orange  #FF4500	Sweep, heat, evidence
WAVEFIELD	Violet  #7B2FBE	Deep purple  #4A0080	Mathematics, abstraction
EMI	Green-gold  #9FE870	Frequency cyan  #00FFD1	Electromagnetic, physics
FORMULA	Racing red  #FF2400	Gold  #FFD700	Competition, achievement

Chapter Entry Mechanic (applies to all 7 chapters)

Before Beat 1 of each chapter, the following always plays:

1. **Locate** — camera drifts toward a bright density node in the supercluster field (2s)
2. **Pulse** — the density node brightens and pulses twice, identifying itself

3. **Extract** — a particle stream peels away from the node and begins moving toward the camera focal point (1s)
4. Beat 1 begins

This 3-4s sequence plays between every chapter. The chapter title fades in on the left as the Extract phase begins.

Chapter Exit Mechanic (applies to all 7 chapters)

Final beat of each chapter always ends:

1. **Pulse** — the entire assembled artifact brightens uniformly (0.5s)
 2. **Shear** — particles accelerate outward from the artifact center (0.8s)
 3. **Rejoin** — sheared particles decelerate and merge into ambient field (1.2s)
 4. Camera pulls back to wide supercluster view and begins travel to next node
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CHAPTER 01 — MIRA

Multilingual Intelligent Real-time Agent Chapter color: Gold #C9A84C Dot count: 8
Catalogue label: MIRA CATALOGUE

DOT 01 — Production System

Proof card

Label: 1/8 Question: *Did this ship?* Answer: **DriveX live leads** Body: Production
outbound voice AI for live DriveX lead conversion. Tags: FASTAPI KUBERNETES

Particle transformation The MIRA density node has been located. Particles in the region pulse gold twice, then the field around the node begins a slow, gentle Condense — particles tightening marginally toward the node center without forming any distinct shape yet. The ambient supercluster continues its drift everywhere else. This is the Locate state: the artifact is implied but not yet assembled.

Camera: Wide. MIRA node at center-right, slightly elevated. Slow drift toward it. The audience is arriving, not yet inside.

Primary behavior: Pulse → early Condense

DOT 02 — Latency Collapse

Proof card

Label: (2/8) Question: *How fast?* Answer: **Sub-100ms** Body: First response dropped from 7s to under 500ms across 4 architectural iterations in 68 days. Tags: (REAL-TIME)

(STREAMING ASR) (FASTAPI)

Particle transformation Long, wide arc filaments form between particle clusters — they are the latency metaphor. These arcs are slow, curved, expensive-looking paths. As this beat settles, the arcs begin contracting: the curved paths shorten and straighten into tight, direct connection lines. The transformation takes the full 1200ms. You watch latency collapse as a physical event — arcs pulling into tight lines.

Camera: Begins moving closer to the node. Slight downward tilt to show the arcs and their contraction in the same frame.

Primary behavior: Route (arc → direct line morph)

DOT 03 — Language Bands

Proof card

Label: (3/8) Question: *Which languages?* Answer: **5 Indian languages** Body: EN, HI, TA, KN, TE — all running in parallel, all live in production. Tags: (EN) (HI) (TA) (KN) (TE)

Particle transformation Five filament bands break from the general field and orbit the forming MIRA core at slightly different radii. Each band carries its own color identity:

- EN: cool white
- HI: saffron ( #FF9933)
- TA: warm red ( #CC2200)
- KN: deep yellow ( #FFD000)
- TE: teal ( #009B8D)

The bands are not rings — they are orbital streams, moving slowly around the core like signal lanes, not planets. The Condense from Dot 02 becomes the attractor they orbit around.

Camera: Rotates 30° clockwise to show the orbital bands from an oblique perspective that reveals their depth and separation.

Primary behavior: Route (orbital stream formation)

DOT 04 — Voice Intake

Proof card

Label: (4/8) Question: *How does voice enter?* Answer: **VAD → ASR → Language**
Body: Three-stage intake pipeline with sub-100ms target per stage. Tags: (SARVAM AI) (STREAMING ASR) (VAD)

Particle transformation A stream of particles enters the artifact from the left — incoming call particles, dense and fast. At a visible split point (approximately 1/3 of the way into the artifact), they divide into three sub-streams:

- VAD stream: bright white, short rapid bursts (voice activity detection — on/off pulses)
- ASR stream: blue-white, continuous flowing filament
- Language stream: color-coded fan — each utterance particle routes toward its matching language band from Dot 03

The fork is the key visual event. The geometry of the three-way split should be clearly readable.

Camera: Rotates to show the intake from slightly above and to the left — the stream entering from the left edge of frame, the fork visible in the center.

Primary behavior: Route (three-way split)

DOT 05 — Orchestration Core

Proof card

Label: 5/8 Question: *Who coordinates it?* Answer: **Pipecat orchestration** Body: Model handoffs, turn management, and agentic state routing across the full call lifecycle. Tags: PIPECAT REDIS CELERY

Particle transformation The split streams from Dot 04 converge into a dense central routing core — the brightest, most particle-dense region in the artifact. Inside the core, particles pulse as they pass through model handoff nodes: a visible brightness event at each handoff point. The core is compact and dense at center, with filaments radiating outward in all directions — connecting intake streams, language bands, and beginning to form output paths. The core Pulses every ~1.5s, synchronized with implied call beats.

Camera: Rotates to center-facing. The routing core occupies the focal center of the frame. Language bands orbit around it. Intake streams arrive from left.

Primary behavior: Condense + Pulse

DOT 06 — Post-Call Output

Proof card

Label: 6/8 Question: *What does it produce?* Answer: **Structured intent** Body:
Confidence score, intent classification, and date extraction — per call, per lead. Tags:
STRUCTURED OUTPUTS ZOH0 CRM DJANGO

Particle transformation Particles exiting the right side of the routing core compress into three distinct tight clusters that hang to the right of the core like structured data objects:

- Intent cluster: densest, gold #C9A84C, roughly spherical
- Confidence cluster: slightly dispersed, softer amber, suggests a probability distribution
- Date/entity cluster: small, tight, grid-arranged particles — the most structured-looking of the three

The three clusters are spatially separated (not merged) so they read as distinct output types.

Camera: Shifts slightly right to frame all three output clusters alongside the routing core. The full artifact is now visible: intake stream left → core center → clusters right.

Primary behavior: Condense (three cluster formation)

DOT 07 — Ops Loop

Proof card

Label: 7/8 Question: *Where does it go?* Answer: **CRM + WhatsApp** Body: Zoho CRM sync and auto-group WhatsApp creation with live bot coordination. Tags: ZOH0
CRM WHATSAPP API FASTAPI

Particle transformation Two long output filaments depart from the cluster group on the right and travel outward into the wider supercluster field — the longest filaments in the MIRA chapter:

- CRM filament: upper-right trajectory, blue-gold color, connects to a distant supercluster density node
- WhatsApp filament: lower-right trajectory, green particles, connects to a different distant node

These filaments cross other ambient particles and filaments in the field — they feel like real infrastructure reaching out into an existing system. The artifact now has its full spatial extent: intake stream on left, core at center, output clusters at right, two long infrastructure filaments departing to the field.

Camera: Pulls back slightly to show the full artifact extent including the outgoing filaments reaching into the wider field.

Primary behavior: Route (long filament departure)

DOT 08 — Ownership

Proof card

Label: 8/8 Question: *Who built it?* Answer: **Sole architect** Body: Architecture, speech pipeline, orchestration, CRM sync, and WhatsApp bot — one engineer, 68

days. Tags: END-TO-END PRODUCTION DEPLOYED

Particle transformation The entire artifact — intake stream, language bands, routing core, output clusters, CRM and WhatsApp filaments — brightens uniformly to peak luminosity. A single pulse radiates outward from the routing core through every connected element. All five language bands brighten simultaneously for the first time. This is the chapter's visual climax.

Then: **Shear → Rejoin**. Particles peel outward from the core periphery and begin returning to the ambient field. The language bands dissolve last. Camera begins travel toward AIDEN.

Primary behavior: Pulse → Shear → Rejoin

CHAPTER 02 — AIDEN

AI-Driven ENgagement Analytics Chapter color: Cyan #00D4FF Dot count: 6
Catalogue label: AIDEN CATALOGUE

DOT 01 — Speaker Separation

Proof card

Label: 1/6 Question: *What is AIDEN?* Answer: **Call intelligence** Body: Full-stack analytics platform giving sales management their first data-driven view into every call. Tags: REACT DJANGO KUBERNETES

Particle transformation Particles arrive from the MIRA Rejoin and from the ambient field. They begin separating into two parallel horizontal bands running left to right — a call timeline orientation:

- Agent band: upper position, cyan ( #00D4FF)
- Customer band: lower position, amber ( #FF9500)

Both bands move left to right continuously, representing call time flowing. The separation between the two bands is the key statement: these are distinct speakers.

Camera: Arrives at AIDEN node from behind. Settles into side-elevation view — both bands running left to right across the frame, clearly separated vertically.

Primary behavior: Condense + Route (two parallel band formation)

DOT 02 — Diarization

Proof card

Label: (2/6) Question: *Can you tell them apart?* Answer: **Speaker diarization** Body:
Every utterance attributed, every speaker separated, across every call in the system.
Tags: (DIARIZATION) (TRANSCRIPTION) (PARALINGUISTICS)

Particle transformation The two parallel bands begin to interweave — particles from each band cross at intervals and braid around each other forming a double helix along the timeline axis. Color distinction is maintained: cyan and amber remain identifiable through the braid. The braid is tight where speakers overlap in conversation, loose where one speaker holds the floor. The helix is the AIDEN artifact's primary shape.

Camera: Rotates to a 30° oblique from the front, revealing the helix depth. The Z-axis of the helix becomes visible.

Primary behavior: Condense (helix formation — the portfolio's most organic shape)

DOT 03 — SOP Scoring

Proof card

Label: (3/6) Question: *How is quality measured?* Answer: **8 SOP dimensions** Body: Greeting, discovery, pitch, objection handling, urgency, close, follow-up, and compliance — scored per call. Tags: (LLM SCORING) (SOP) (ANALYTICS)

Particle transformation Eight thin horizontal particle planes slice through the helix at evenly-spaced intervals along the timeline axis. The planes are semi-transparent so the helix is visible through them. Each plane Pulses at the intersection point with the helix — a bright event showing the evaluation moment. The planes activate sequentially as the scroll settles, not simultaneously — giving the impression of a system checking each dimension.

Camera: Slight overhead tilt to show the full helix with all 8 scoring planes visible simultaneously from above.

Primary behavior: Pulse (planes activating sequentially)

DOT 04 — Parallel LLM Tracks

Proof card

Label: (4/6) Question: *How deep is the analysis?* Answer: **Parallel LLM analysis** Body: Multiple models run simultaneously across emotion, content, and SOP compliance — not sequential. Tags: (PARALLEL LLM) (HUME AI) (STRUCTURED OUTPUTS)

Particle transformation Three secondary filament tracks split from the main helix — parallel streams running alongside it at the same velocity. Each track carries particles at different densities:

- Emotion track: fastest moving, warmest color

- Content track: medium speed, cooler
- Compliance track: slowest, most structured

At four points along the timeline, each parallel track rejoins the helix at insight nodes — bright convergence points where the LLM output fuses back into the main conversation structure. The artifact now has lateral complexity: one primary helix plus three analysis tracks running parallel.

Camera: Pulls back to show the full width of the parallel structure — helix at center, three tracks flanking.

Primary behavior: Route (parallel track formation + rejoin at insight nodes)

DOT 05 — Dashboard Surface

Proof card

Label: 5/6 Question: *Where does it surface?* Answer: **Zoho CRM embedded** Body: React dashboard embedded inside Zoho CRM — sales management visibility without changing tools. Tags: REACT ZOHO CRM EMBEDDED

Particle transformation Particles departing the right end of the helix (post-call output) organize into a three-dimensional particle grid — the dashboard surface. Not a flat card but a slightly angled, layered structure with rows and columns suggesting data panels with depth. The grid floats to the right of the helix, connected by a short filament bridge. The grid is distinct from the helix's organic shape — it is structured, rectilinear, deliberate.

Camera: Shifts right to frame the helix-to-dashboard flow in one composition. The organic helix on left, the structured grid on right, the filament bridge between.

Primary behavior: Condense (particles from organic filaments into structured grid)

DOT 06 — Reliability

Proof card

Label: 6/6 Question: *Does it hold?* Answer: **K8s + E2E coverage** Body: Production-grade Kubernetes deployment with full end-to-end test coverage. Shipped in 31 days.
Tags: KUBERNETES PLAYWRIGHT E2E

Particle transformation A perimeter of green-gold particles forms around the entire artifact — helix, parallel tracks, and dashboard grid. The perimeter is not a solid shell: it is a distributed guardrail of particle clusters at regular intervals connected by thin filaments, forming a sparse but complete enclosure. Inside the enclosure, the artifact brightens. The guardrail represents deployment confidence — the system is contained, tested, and running.

Then: **Pulse → Shear → Rejoin**. Camera begins travel toward VANGUARD node.

Primary behavior: Condense (guardrail formation) → Pulse → Shear → Rejoin

CHAPTER 03 — VANGUARD

Autonomous Web Testing Agent Chapter color: Red-violet #8B2FC9 (fail) + Pass-green #00FF7F (pass) Dot count: 5 Catalogue label: VANGUARD CATALOGUE

DOT 01 — Browser Graph

Proof card

Label: 1/5 Question: *What does Vanguard do?* Answer: **Autonomous web testing**
Body: Validates the DriveX corporate website end-to-end without a single scripted

test case. Tags: VLM PLAYWRIGHT AUTONOMOUS

Particle transformation Particles condense into a directed acyclic graph viewed at isometric angle — nodes are browser states (individual pages), edges are navigation paths. The graph has clear hierarchy: root node at top center, branching downward and outward. Node particles are bright and compact. Edge particles are thinner, directional (with a subtle flow indicating navigation direction). The graph occupies the central frame and reads immediately as a web of connected states.

Camera: Arrives at isometric angle, slightly elevated. The full DAG is visible. No movement — the structure establishes itself while the camera holds.

Primary behavior: Condense (DAG formation)

DOT 02 — Agent Probe

Proof card

Label: 2/5 Question: *How does it test?* Answer: **VLM + Playwright** Body: A local vision model observes the page state and decides the next action — no scripts, no selectors. Tags: VLM PLAYWRIGHT AGENTIC

Particle transformation A single particle — significantly brighter than all others, gold — enters the graph at the root node and begins traversing paths. It leaves a luminous trail as it moves, so its navigation history is visible. The probe pauses at each decision node for ~300ms (processing), brightens, then commits to a direction. The agent's autonomy is shown by the deliberate pause before each decision.

Camera: Follows the probe at medium distance. Slight drift tracking its movement through the graph.

Primary behavior: Route (probe traversal with luminous trail)

Proof card

Label: (3/5) Question: *What happens when it fails?* Answer: **Reroute and document**

Body: Failed paths detected, logged as red-violet, and the agent finds an alternate route autonomously. Tags: (FAILURE DETECTION) (RETRY LOGIC) (LOGGING)

Particle transformation Three nodes in the graph flash red-violet (■ #8B2FC9) and their connecting edges shift color to match — failed paths. The probe approaches a failed node, pauses longer than usual (500ms), then reverses along its luminous trail and finds an alternate route. The alternate route glows green (■ #00FF7F) as the probe traverses it. The failed nodes remain red-violet permanently — they are documented, not erased. The contrast between red-violet (documented failure) and green (successful reroute) is the chapter's visual thesis.

Camera: Stays with the probe. Slight push-in when the failure flash occurs to emphasize the moment.

Primary behavior: Pulse (red-violet failure flash) → Route (reroute in green)

DOT 04 — DOM Diagnostic Field

Proof card

Label: (4/5) Question: *What does it see?* Answer: **DOM + screenshot** Body: The agent reads both structural state and visual state simultaneously — not one or the other.

Tags: (DOM INSPECTION) (VISUAL DIFF) (PLAYWRIGHT)

Particle transformation Each graph node expands slightly and reveals a sub-lattice of finer particles — the DOM tree structure inside each page. The sub-lattices are denser and more granular than the main graph edges. Behind each node, a flat plane of particles appears at a slight recline — the screenshot capture surface, like a visual backing to the

structural lattice. The graph now has three visible depth layers: graph topology in front, DOM sub-lattice in middle, screenshot plane behind.

Camera: Pushes in toward one node to show the DOM sublattice and screenshot plane in close detail. Then pulls back to show the full multi-layer structure from the isometric angle.

Primary behavior: Condense (sub-lattice emergence within each node)

DOT 05 — Release Risk Map

Proof card

Label: 5/5 Question: *What's the output?* Answer: **Release confidence** Body: A continuous risk signal across the full site — not a pass/fail checklist. Tags: RISK
SIGNAL CONTINUOUS QA

Particle transformation The entire graph undergoes a color temperature shift across all nodes and edges simultaneously:

- Green nodes: low-risk (consistent pass state)
- Amber nodes: medium-risk (intermittent failures logged)
- Red-violet nodes: high-risk (persistent failure pattern)

The graph reads as a heat map of deployment confidence — the same structure from Dot 01, now carrying meaning in its colors. The probe particle rests at the root node. The artifact holds for 1.5s as a complete risk portrait.

Then: **Pulse → Shear → Rejoin**. Camera travels toward INSPECTION node.

Primary behavior: Pulse (color temperature shift) → Shear → Rejoin

CHAPTER 04 — INSPECTION

AI-Powered Vehicle Inspection System Chapter color: Scan blue (#A8D8FF) (rays) + Defect orange (#FF4500) (heat zones) Dot count: 5 Catalogue label: INSPECTION CATALOGUE

DOT 01 — Silhouette

Proof card

Label: (1/5) Question: *What does it inspect?* Answer: **1,000+ vehicle parts** Body: AI-powered inspection of two-wheeler vehicles across the full parts inventory. Tags: (YOLOV8) (COMPUTER VISION) (ELECTRON)

Particle transformation Particles carve a two-wheeler silhouette — a scooter or motorcycle — from negative space. Particle density concentrates at the perimeter: frame, wheel arches, body panels, handlebars, engine casing outline. The interior is void — no particles inside the silhouette, only at its edges. The shape is legible from a 3/4 front angle as a two-wheeler. No solid mesh. Pure boundary density. This is the portfolio's defining Carve behavior.

Camera: 3/4 front angle, medium distance. The silhouette fills the center of frame. Holds steady while the Carve completes.

Primary behavior: Carve (the portfolio's defining example of this behavior)

DOT 02 — Camera Sweep

Proof card

Label: (2/5) Question: *How does it find defects?* Answer: **YOLOv8 detection** Body: Computer vision model running inference across live RTSP camera feeds from inspection stations. Tags: (YOLOV8) (RTSP) (REAL-TIME)

Particle transformation Four ice-blue ray filaments emerge from fixed points around the vehicle silhouette (representing four physical cameras) and sweep across the surface simultaneously. Each ray moves from one side to the other at a constant speed. Where a ray crosses a normal surface zone: particles briefly brighten blue-white and return to neutral. Where a ray crosses a defect zone: particles heat from blue-white to orange-red (● #FF4500) and hold their temperature. Three defect zones glow orange-red after the sweep completes — one on the front panel, one on the rear wheel arch, one on the body.

Camera: Holds 3/4 angle while rays sweep. Slight clockwise rotation during sweep to show rays arriving from different physical positions.

Primary behavior: Route (sweep rays) → Pulse (defect zones hold heat)

DOT 03 — Live Streams

Proof card

Label: (3/5) Question: *What's the source?* Answer: **Live RTSP feeds** Body: Real-time camera streams from physical inspection stations in DriveX showrooms. Tags: (RTSP) (SHOWROOM) (REAL-TIME)

Particle transformation Four thin horizontal particle bands appear in the field behind the vehicle silhouette at different depths — the four RTSP stream feeds represented as parallel data flows. Each band moves left to right continuously at slightly different speeds, creating a sense of live, asynchronous data arrival. The bands do not interfere with the vehicle silhouette — they pass behind it. The scene now has foreground (vehicle + defect zones) and background (stream bands).

Camera: Pulls back slightly to show the vehicle in context with the stream bands visible behind it.

Primary behavior: Route (continuous background stream movement)

DOT 04 — Operations Output

Proof card

Label: 4/5 Question: *Where does it run?* Answer: **Electron desktop app** Body: Showroom inspection tool with live feed monitoring, report generation, and part-level history. Tags: ELECTRON REACT STRAPI

Particle transformation Particles from the right side of the vehicle and from the defect zones begin organizing into two output surfaces floating beside the vehicle at a slight angle:

- Upper surface: Electron desktop panel — rectangular flat particle grid, slightly elevated, suggestions of a window frame at its edges
- Lower surface: Report panel — denser, more compact, suggesting structured data output

Both surfaces connect to the vehicle by short filaments. They are abstract — readable as screens/interfaces without rendering actual UI elements.

Camera: Shifts right to frame the vehicle alongside the two output surfaces in one composition.

Primary behavior: Condense (filament particles organizing into panel grids)

DOT 05 — Inspection Report

Proof card

Label: (5/5) Question: *What's the output?* Answer: **Structured evidence** Body: Consistent inspection report per vehicle — defect location, severity, and image reference, every time. Tags: (REPORT) (CONSISTENT) (STRUCTURED)

Particle transformation From the defect zones on the vehicle and the output panels, a structured column of particles departs downward-right. The column is ordered — groups of particles representing each defect finding travel in formation, not scatter. The groups are evenly spaced, suggesting a line-item report structure. As the column departs, the vehicle silhouette dims gradually. The orange-red defect zones are the last elements to remain bright before Shearing.

Then: **Shear** → **Rejoin**. Camera travels toward WAVEFIELD node.

Primary behavior: Route (structured report departure) → Shear → Rejoin

CHAPTER 05 — WAVEFIELD

Sub-Quadratic Attention Mechanism Research Chapter color: Violet (#7B2FBE) Dot count: 5 Catalogue label: WAVEFIELD CATALOGUE

DOT 01 — Attention Wall

Proof card

Label: (1/5) Question: *What's the problem?* Answer: **O(n²) attention** Body: Standard transformer attention scales quadratically — at 1M tokens, it becomes

computationally impossible. Tags: TRANSFORMER ATTENTION COMPLEXITY

Particle transformation Particles arrange into a large triangular matrix — rows of token particles across the top and down the left side. Every pair of tokens connects with a thin violet filament. The upper triangle of the matrix is filled with these connection lines to near-opacity — so dense the individual filaments are barely distinguishable. The visual impression is intentional: heavy, congested, expensive. A wall of computation.

Camera: Top-down, slight forward tilt. The triangular matrix fills the frame. Should feel oppressive in its density. Hold — don't move.

Primary behavior: Condense (matrix locks in; density is the visual statement)

DOT 02 — Wave Transform

Proof card

Label: 2/5 Question: *What's the approach?* Answer: **Wave field decomposition**

Body: Replacing pairwise dot-product attention with Fourier-based wave functions and Green's function paths. Tags: FOURIER GREEN FUNCTION WAVE FIELD

Particle transformation This is the most dramatic transformation in the portfolio. The matrix Shears — the dense connection filaments dissolve simultaneously. The token rows begin to ripple, and the matrix surface folds: particles that were discrete connection lines become a continuous undulating wave surface. The flat heavy matrix becomes a three-dimensional wave field. Fourier bands run horizontally across the wave surface at logarithmic spacing — the mathematical structure of the mechanism.

The transformation should take 1500ms — longer than any other single beat. The dissolution of the matrix and the emergence of the wave happen in one continuous morph, not two sequential steps.

Camera: Begins overhead during the matrix, tilts to 45° during the transformation to reveal the wave surface depth as it forms.

Primary behavior: Shear (matrix dissolves) → Condense (wave surface forms)

DOT 03 — Content Gates

Proof card

Label: $\boxed{3/5}$ Question: *How does it stay selective?* Answer: **Content-dependent gating** Body: The mechanism attends to relevant tokens without computing all pairs — selectivity without quadratic cost. Tags: $\boxed{\text{GATING}}$ $\boxed{\text{SELECTIVE ATTENTION}}$ $\boxed{\text{KERNEL}}$

Particle transformation Vertical particle slices appear across the wave surface — gates that open and close. When a gate opens, the wave particles behind it brighten and a connection is active. When closed, those particles dim and the connection is suppressed. Multiple gates cycle open and closed across the wave surface in a pattern, creating visible selective attention — most of the surface is dimmed, a few gates are open at any time. Sparse, deliberate, efficient.

Camera: Holds at 45° angle. Gates visible across the full wave width. Camera does not move — let the gates animate.

Primary behavior: Pulse (gates opening and closing)

DOT 04 — Complexity Compression

Proof card

Label: $\boxed{4/5}$ Question: *What's the result?* Answer: **$O(n \log n)$** Body: $\sim 200,000\times$ theoretical speedup at 1M tokens, grounded in the Alman-Yu lower bound (ICLR 2025). Tags: $\boxed{O(N \log N)}$ $\boxed{\text{ALMAN-YU}}$ $\boxed{\text{ICLR 2025}}$

Particle transformation A single bright route line traces itself along the wave crest from left to right — the $O(n \log n)$ complexity path. It follows the wave's natural contour: not crossing every particle pair, but traveling efficiently along the surface. On the left side of the wave, a faint ghost of the original triangular matrix density remains — barely visible, faded. On the right, the route line is bright and clean. The contrast says everything: dense matrix on left ghost, elegant route line on right.

Camera: Moves alongside the route line from left to right — the camera's own movement illustrating linear traversal through the problem.

Primary behavior: Route (complexity line traces itself)

DOT 05 — Research Framework

Proof card

Label: 5/5 Question: *What's the deliverable?* Answer: **Complete research paper**
Body: Derived mechanism, theoretical grounding in Alman-Yu lower bound, complexity proof, and full paper. Tags: RESEARCH THEORY PAPER

Particle transformation The wave surface settles into its final form. Particles at key structural points — gate positions, the crest route line, the Fourier band intersections, the boundary conditions — brighten and hold. Surrounding wave particles dim to a soft ambient glow. The artifact is now a research diagram: a documented wave surface with marked proof points. It looks like mathematics, not a product.

Then: **Pulse → Shear → Rejoin**. Camera travels toward EMI node.

Primary behavior: Pulse (proof points hold) → Shear → Rejoin

CHAPTER 06 — EMI

Electromagnetic Interference Shielding Simulator Chapter color: Green-gold

 #9FE870 Dot count: 5 Catalogue label: EMI CATALOGUE

DOT 01 — Material Layers

Proof card

Label:  Question: *What's being modelled?* Answer: **Multi-phase shielding** Body:
Composite EMI shielding simulation across multiple material layers with real
microstructure composition. Tags:   

Particle transformation Particles form a horizontal half-pipe cross-section — a tube cut lengthwise and viewed from the open side, oriented horizontally in space. Inside, concentric ring layers appear one at a time from the inside outward:

- Innermost ring: densest, brightest — substrate
- Middle rings: decreasing density — composite layers
- Outermost ring: lightest, most diffuse — surface coating

Each ring has a slightly different particle density and green-gold color temperature. The structure is clearly a material cross-section diagram.

Camera: Side elevation with slight elevation angle. The concentric ring cross-section is the primary read.

Primary behavior: Condense (layers build inside-out)

Proof card

Label: **2/5** Question: *What's the test range?* Answer: **100 kHz to 10 GHz** Body: Full frequency sweep across the material cross-section — from low-frequency to microwave range. Tags: **FREQUENCY SWEEP** **100KHZ** **10GHZ**

Particle transformation From the left opening of the half-pipe, a wavefront enters — a dense ring of particles representing the incoming electromagnetic wave. It travels rightward through the concentric material layers at a steady pace. As it crosses each layer:

- Some particles scatter backward (leftward scatter — reflection)
- Some particles disappear mid-layer (absorption)
- Remaining particles continue rightward at reduced density (transmission)

The wavefront arriving at the right end is significantly attenuated — visibly thinner and dimmer than the entering wavefront.

Camera: Holds side elevation to track the wavefront traveling left to right across the full cross-section.

Primary behavior: Route (wavefront traversal with attenuation)

DOT 03 — Output Curves

Proof card

Label: **3/5** Question: *What does it measure?* Answer: **Reflection + Absorption + SE** Body: Three separated output terms — reflection loss, absorption loss, and total shielding effectiveness. Tags: **SHIELDING EFFECTIVENESS** **dB** **SCHUNKOFF**

Particle transformation On the right end of the half-pipe, particles organize into three separated upward-rising curves:

- Reflection curve: rises sharply early, plateaus — leftmost
- Absorption curve: rises steadily across the range — center
- Total SE curve: rises highest and holds — rightmost

The three curves are vertically separated so they are clearly distinct, but all emerge from the same right-end source point. Each curve is made of particles at the measurement points — not a continuous line but a particle-sampled curve shape.

Camera: Shifts right to frame the three output curves alongside the material cross-section.

Primary behavior: Condense (scattered exit particles organizing into curve shapes)

DOT 04 — Microstructure

Proof card

Label: 4/5 Question: *What makes it accurate?* Answer: **Real microstructure** Body: Composite material composition modelled at the microstructure level — not idealized homogeneous layers. Tags: MICROSTRUCTURE COMPOSITE VECTORISED

Particle transformation Inside each material ring layer, fine irregular particle clusters appear — distributed non-uniformly within each ring, representing the actual composite microstructure. The rings are no longer clean and uniform — they have texture, variance, grain. These microstructure points Pulse gently at irregular intervals. Each pulse event on a microstructure point causes a small corresponding shift in the three output curves on the right — the curves respond in real time to the material's internal structure.

Camera: Pushes in close to one material layer to show the microstructure texture. Holds, then pulls back to show the full system with curves responding.

Primary behavior: Pulse (microstructure points) → Route (curve response)

DOT 05 — Validation Envelope

Proof card

Label: 5/5 Question: *What's the deliverable?* Answer: **Validation envelope** Body: Frequency-dependent shielding effectiveness bounds with 20-year service life prediction. Tags: VALIDATION SERVICE LIFE JENKINS

Particle transformation A particle envelope forms around the three output curves — an upper bound and lower bound curving around the central cluster of curves. The envelope is wider at the frequency extremes (more uncertainty) and tighter in the validated center range (high confidence). The envelope represents the simulation's confidence bounds. The system dims slightly outside the envelope region — the validated range is the deliverable.

Then: **Pulse → Shear → Rejoin**. Camera travels toward FORMULA node.

Primary behavior: Condense (envelope formation) → Shear → Rejoin

CHAPTER 07 — FORMULA

Formula Manipal Race Operations Chapter color: Racing red ■ #FF2400 + Gold ■ #FFD700 Dot count: 5 Catalogue label: FORMULA CATALOGUE

Proof card

Label: (1/5) Question: *What's the context?* Answer: **Formula Manipal** Body:

Operations Lead for the FM23e electric race car. Formula Bharat 2024. Tags: (FORMULA

(BHARAT) (EV) (AUTONOMOUS)

Particle transformation Particles form a top-down circuit layout — a single looping filament tracing a race track shape: straights, hairpins, chicanes, a pit lane entry. The track path traces itself from the start/finish line around the full circuit and returns to origin. No buildings, no car yet — just the track as a closed particle loop. The track path is bright racing red.

Camera: High top-down. Track fills the frame. Slow clockwise rotation over 3s to reveal the full circuit shape.

Primary behavior: Route (track filament traces itself — start to finish to start)

DOT 02 — Telemetry Stream

Proof card

Label: (2/5) Question: *What did the car produce?* Answer: **Live telemetry** Body: Real-time data capture from track-side testing sessions driving path planning

improvements. Tags: (TELEMETRY) (PATH PLANNING) (TRACK-SIDE)

Particle transformation A vehicle particle — distinctly bright, tight cluster — enters the circuit at the start line and begins lapping. Behind it, a ribbon of telemetry particles trails:

- Red (● #FF2400): high speed sections
- Amber: braking zones

- Green: apex and cornering

The ribbon fades after $\sim 180^\circ$ of track — only the recent half-lap of telemetry is visible at any time, giving it a live-data feel. The car particle continues lapping continuously.

Camera: Lowers from high top-down to a 40° perspective angle that shows the circuit and the telemetry ribbon with depth.

Primary behavior: Route (car traversal + trailing telemetry ribbon)

DOT 03 — Racing Line

Proof card

Label: 3/5 Question: *What improved?* Answer: **40% control accuracy** Body: Autonomous path planning improvements validated through track-side testing — tighter lines, earlier power. Tags: PATH PLANNING AUTONOMOUS 40% IMPROVEMENT

Particle transformation Two racing lines appear on the circuit simultaneously:

- Original line: dim blue — wider entry into corners, conservative apex, late power application
- Improved line: bright gold ☐ #FFD700 — tighter entries, earlier apex, visible earlier power application zone

The car particle moves along the gold line. At each corner, the gap between the two lines is visible and measurable — the distance between them is the proof of improvement. The improvement is largest at the hairpin corners.

Camera: Tracks through two corners at a low angle — 20° from track level — showing the line differential from behind the car.

Primary behavior: Pulse (gold line brightens over dim blue at each apex)

DOT 04 — Operations Network

Proof card

Label: 4/5 Question: *What else?* Answer: **₹60L in sponsorships** Body: Operations leadership across engineering, strategy, sponsorship, manufacturing, and competition logistics. Tags: OPERATIONS SPONSORSHIP MANUFACTURING

Particle transformation Six operation node clusters appear around the track perimeter — outside the circuit boundary, at even intervals:

- Engineering node: upper left
- Strategy node: upper right
- Pit operations node: near pit lane
- Sponsorship node: lower right
- Manufacturing node: lower left
- Media node: far right

Each node is a compact bright particle cluster. Thin gold filaments connect every node to the moving car particle and to each adjacent node — forming a complete operations network around the racing artifact. The track circuit now has an organizational system surrounding it.

Camera: Pulls back to wide top-down to show the full system: track + car + telemetry + operations network as one connected structure.

Primary behavior: Route (network filaments connect all nodes to car and to each other)

DOT 05 — Competition Constellation

Proof card

Label: (5/5) Question: *What's the result?* Answer: **1st place, Cost & Manufacturing**
Body: Formula Bharat 2024. The full operation — path planning, trackside testing, sponsorship, manufacturing — delivered. Tags: (FORMULA BHARAT 2024) (1ST PLACE)
(DELIVERED)

Particle transformation All elements pulse simultaneously — track, car, telemetry ribbon, racing lines, operations nodes, network filaments — in a single bright event. The six operations nodes brighten in rapid sequence (achievement cascade). The car particle completes its current lap and arrives back at the start/finish line, which brightens as a finish line marker. The entire artifact — track plus surrounding network — holds as a constellation for 2s.

Then: the largest Shear in the portfolio. Every particle departs outward with more velocity than any previous chapter, because this is the final chapter release. The Rejoin brings all FORMULA particles — and trails of every previous chapter — back into the full supercluster simultaneously.

Primary behavior: Pulse (achievement cascade) → Shear (final, widest) → Rejoin

Closing State — The Restored Supercluster

After the FORMULA Rejoin, the full supercluster is restored — but different from the opening.

What has changed:

- All seven chapter nodes are now active (previously only the path to MIRA was illuminated)
- All seven route lines connecting the nodes are simultaneously visible

- The ambient field carries traces of all chapter colors: gold, cyan, red-violet, orange, violet, green-gold, racing red — mixed into the base cobalt field

Proof card (final):

┃ No chapter label No question Answer: **Seven systems. One engineer.** No tags

The camera holds the restored supercluster in the same wide view as the portfolio opening. The route lines between the seven nodes are the final image — a body of connected work, not a list of jobs.

No Shear. No Rejoin. The field stays alive.



Dot Pagination Reference

Chapter	Dots	Beat Names
MIRA	8	Production System · Latency Collapse · Language Bands · Voice Intake · Orchestration Core · Post-Call Output · Ops Loop · Ownership
AIDEN	6	Speaker Separation · Diarization · SOP Scoring · Parallel LLM · Dashboard Surface · Reliability
VANGUARD	5	Browser Graph · Agent Probe · Fail and Reroute · DOM Diagnostic · Release Risk Map
INSPECTION	5	Silhouette · Camera Sweep · Live Streams · Operations Output · Inspection Report
WAVEFIELD	5	Attention Wall · Wave Transform · Content Gates · Complexity Compression · Research Framework

Chapter	Dots	Beat Names
EMI	5	Material Layers · Frequency Sweep · Output Curves · Microstructure · Validation Envelope
FORMULA	5	Track Path · Telemetry Stream · Racing Line · Operations Network · Competition Constellation
Total	39	

Particle Count Guidelines

Chapter	Chapter Particle Pool	Behavior Complexity	Most Demanding Beat
MIRA	~15% of field	High	Dot 04 (three-way split)
AIDEN	~12% of field	High	Dot 02 (helix formation)
VANGUARD	~10% of field	Medium	Dot 03 (probe reroute)
INSPECTION	~8% of field	Medium	Dot 01 (Carve silhouette)
WAVEFIELD	~18% of field	Very High	Dot 02 (matrix → wave morph)
EMI	~10% of field	Medium	Dot 02 (wavefront traversal)
FORMULA	~12% of field	Medium	Dot 04 (full network formation)

WAVEFIELD Dot 02 (the matrix-to-wave morph) is the single most computationally demanding moment in the portfolio. If performance budgeting is required, this is the beat to optimize first.

Quality Gate Checklist

Before any chapter goes to implementation:

- ☐ Artifact shape is legible from the designated camera angle without labels
- ☐ Each beat shows visible particle movement — no beat is a static state
- ☐ The proof claim is visible in the particle state, not only in the card text
- ☐ Chapter color is traceable across all beats
- ☐ Shear + Rejoin plays completely before camera begins travel
- ☐ Ambient field particles remain active in background throughout all beats
- ☐ WAVEFIELD Dot 02 morph duration is $\geq 1500\text{ms}$
- ☐ INSPECTION Dot 01 Carve reads as a two-wheeler from the designated 3/4 front angle